

TASK-05

Demonstrate how to pull a Docker image, tag it, push it to Docker Hub, and create a new image by installing software inside a running container. Also verify the installed software and save the changes using Docker commit.

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PS C:\Users\VICTUS> docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
failed to do request: Get "https://registry-1.docker.io/v2/library/alpine/referers/sha256:fe2385f276937dcf780967a5385767fd34b34580c8ed8d303a0cd1485a692635": tls: failed to verify certificate: x509: certificate signed by unknown authority
PS C:\Users\VICTUS> docker tag alpine kavya4619m/ap2:v2.1
PS C:\Users\VICTUS> docker login
Authenticating with existing credentials... [Username: kavya4619m]

Info →To login with a different account, run 'docker logout' followed by 'docker login'

Login did not succeed, error: Error response from daemon: Get "https://registry-1.docker.io/v2/": tls: failed to verify certificate: x509: certificate signed by unknown authority

USING WEB-BASED LOGIN

Info →To sign in with credentials on the command line, use 'docker login -u <username>'
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Your one-time device confirmation code is: TFJX-QDHT
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
Post "https://hub.docker.com/v2/access-tokens/desktop-generate": tls: failed to verify certificate: x509: certificate signed by unknown authority

PS C:\Users\VICTUS> docker run -it --name a1 kavya4619m/ap2:v2.1
/ # apt update
/bin/sh: apt: not found
/ # apt add php
/bin/sh: apt: not found
/ # apk update
[fetch https://dl-cdn.alpinelinux.org/alpine/v3.23/community/x86_64/APKINDEX.tar.gz]
v3.23.3-84-gbacb23b6fa2 [https://dl-cdn.alpinelinux.org/alpine/v3.23/main]
v3.23.3-85-g9c9ed52dic6 [https://dl-cdn.alpinelinux.org/alpine/v3.23/community]
OK: 27572 distinct packages available
/ #
/ # apk add php
(1/9) Installing php84-common (8.4.17-r0)
ERROR: php84-common-8.4.17-r0: TLS: server certificate not trusted
(2/9) Installing argon2-libs (20190702-r5)
```

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v3.23.3-85-g9c9ed52dic6 [https://dl-cdn.alpinelinux.org/alpine/v3.23/community]
OK: 27572 distinct packages available
/ # apk add php
(1/9) Installing php84-common (8.4.17-r0)
(2/9) Installing argon2-libs (20190702-r5)
(3/9) Installing ncurses-terminfo-base (6.5_p20251123-r0)
(4/9) Installing libncursesw (6.5_p20251123-r0)
(5/9) Installing libedit (20251016.3.1-r0)
(6/9) Installing pcre2 (10.47-r0)
(7/9) Installing xz-libs (5.8.2-r0)
(8/9) Installing libxml2 (2.13.9-r0)
(9/9) Installing php84 (8.4.17-r0)
Executing busybox-1.37.0-r30.trigger
OK: 18.7 MiB in 25 packages
/ # php -v
PHP 8.4.17 (cli) (built: Jan 20 2026 19:04:32) (NTS)
Copyright (c) The PHP Group
Built by Alpine Linux aports
Zend Engine v4.4.17, Copyright (c) Zend Technologies
/ # exit
PS C:\Users\VICTUS> docker commit a2 kavya4619m/ap2:v2.1
sha256:69eae0896851178945f19e56d4a1e2ea54153e5909eedca78b29949488fddb6
PS C:\Users\VICTUS>
```

```
Executing busybox-1.37.0-r30.trigger
OK: 18.7 MiB in 25 packages
/ # php -v
PHP 8.4.17 (cli) (built: Jan 20 2026 19:04:32) (NTS)
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/ # exit
PS C:\Users\VICTUS> docker commit a2 kavya4619m/ap2:v2.1
sha256:69eae0896851178945f19e56d4a1e2ea54153e5909eedca78b29949488fddb6
PS C:\Users\VICTUS> docker stop a2
a2
PS C:\Users\VICTUS> docker rm a2
a2
PS C:\Users\VICTUS> docker run -it --name a3 kavya4619m/ap3:v3.1
Unable to find image 'kavya4619m/ap3:v3.1' locally
PS C:\Users\VICTUS> docker run -it --name a3 kavya4619m/ap2:v2.1
/ # php -v
PHP 8.4.17 (cli) (built: Jan 20 2026 19:04:32) (NTS)
Copyright (c) The PHP Group
Built by Alpine Linux aports
Zend Engine v4.4.17, Copyright (c) Zend Technologies
/ #
```

```
Built by Alpine Linux aports
Zend Engine v4.4.17, Copyright (c) Zend Technologies
/ # exit
PS C:\Users\VICTUS> docker commit a2 kavya4619m/ap2:v2.1
sha256:69eae0896851178945f19e56d4a1e2ea54153e5909eedca78b29949488fddb6
PS C:\Users\VICTUS> docker stop a2
a2
PS C:\Users\VICTUS> docker rm a2
a2
PS C:\Users\VICTUS> docker run -it --name a3 kavya4619m/ap3:v3.1
Unable to find image 'kavya4619m/ap3:v3.1' locally
PS C:\Users\VICTUS> docker run -it --name a3 kavya4619m/ap2:v2.1
/ # php -v
PHP 8.4.17 (cli) (built: Jan 20 2026 19:04:32) (NTS)
Copyright (c) The PHP Group
Built by Alpine Linux aports
Zend Engine v4.4.17, Copyright (c) Zend Technologies
/ # exit
PS C:\Users\VICTUS> docker push kavya4619m/ap2:v2.1
The push refers to repository [docker.io/kavya4619m/ap2]
589002ba0eae: Mounted from library/alpine
dd6f5da2f700: Pushing [=====>] 6.268MB/6.268MB
```

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Windows PowerShell
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PS C:\Users\VICTUS> mkdir dockerfile_prep_1

Directory: C:\Users\VICTUS

Mode                LastWriteTime         Length Name
----                -
d-----         12-02-2026   12:31 PM             dockerfile_prep_1

PS C:\Users\VICTUS> cd dockerfile_prep_1
PS C:\Users\VICTUS\dockerfile_prep_1> notepad dockerfile
PS C:\Users\VICTUS\dockerfile_prep_1> docker build -t ub1 .
[+] Building 0.2s (1/1) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 2B                                0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory
PS C:\Users\VICTUS\dockerfile_prep_1> ren dockerfile.txt dockerfile
PS C:\Users\VICTUS\dockerfile_prep_1> dir
RAM 0.85 GB  CPU 0.08%  Disk: 5.01 GB used (limit 1006.85 GB)
```

```
Directory: C:\Users\VICTUS\dockerfile_prep_1

Mode                LastWriteTime         Length Name
----                -
-a-----         12-02-2026   12:36 PM             223 dockerfile

PS C:\Users\VICTUS\dockerfile_prep_1> docker build -t ub1 .
[+] Building 155.2s (6/6) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile              0.0s
=> => transferring dockerfile: 262B                                0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.2s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/2] FROM docker.io/library/ubuntu:latest@sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b 0.1s
=> => resolve docker.io/library/ubuntu:latest@sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b 0.0s
httpd:latest          dd178595edd6         177MB         47.6MB      U
kavya4619n/ap1:v1.1   25109184c71b         13.1MB        3.95MB      U
kavya4619n/ap2:v2.1   69eae0896851         33.6MB        10.1MB      U
mongo:latest          6f55a078e025         1.27GB        333MB       U
mysql:latest          6b18d01fb632         1.27GB        283MB       U
nginx:latest          c881927c4077         240MB         65.7MB      U
RAM 0.86 GB  CPU 0.17%  Disk: 5.01 GB used (limit 1006.85 GB)
```

```
httpd:latest          dd178595edd6         177MB         47.6MB      U
kavya4619n/ap1:v1.1   25109184c71b         13.1MB        3.95MB      U
kavya4619n/ap2:v2.1   69eae0896851         33.6MB        10.1MB      U
mongo:latest          6f55a078e025         1.27GB        333MB       U
mysql:latest          6b18d01fb632         1.27GB        283MB       U
nginx:latest          c881927c4077         240MB         65.7MB      U
ub1:latest            a423edda30b1         412MB         115MB       U
ubuntu:latest         cd1dba651b30         119MB         31.7MB      U
PS C:\Users\VICTUS\dockerfile_prep_1> docker run -it ub1:latest
Hello Students
PS C:\Users\VICTUS\dockerfile_prep_1> 
```

- Step 1: Pull Alpine Linux Image
- Step 2: Tag the Alpine Image
- Step 3: Login to Docker Hub
- Step 4: Push the Image to Docker Hub
- Step 5: Run the Image Interactively

Step 6: Update Package Index Inside Container

Step 7: Install PHP Inside Container

Step 8: Verify PHP Installation

Step 9: Exit the Container

Step 10: Commit Container Changes to a New Image